The Global Fertility Dividend: Smaller Families and the Fall in Absolute Poverty

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Abstract

Global poverty is measured using household resources per person. When family size falls, resources per person rise mechanically, reducing poverty and causing demographic change to be recorded as income growth. I decompose the contribution of this "fertility dividend" with rank-preserving transports of the distribution of co-resident children per mother across adjacent surveys in 22 countries that together held 81% of the world's \$3.00-per-day poor in 1990. On a fixed sample that includes countries where poverty rose, smaller families account for 14.0% of the all-person decline at \$3.00 per day, 18.8% of the child-poverty decline, and 16.7% of the decline among people in households with children—a cumulative contribution of 160 million people. The analysis is robust to alternative rank assignments and equivalence scales.

Keywords: Poverty; fertility; family size; equivalence scales; demographic change
JEL Classification: I32, J13, O15

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1 Introduction

Global absolute poverty fell dramatically over the past forty years. A standard accounting credits this decline to growth in household resources and to changes in their distribution. This paper assesses the contribution of demographic change from falling fertility. As families had fewer children, household resources were spread across fewer people, reducing measured poverty even with resources held fixed. This direct resource-dilution channel is the microeconomic analog of a fertility dividend.

This dividend is largely absent from poverty accounting. Poverty measures typically condition on family size, and often convert household resources into resources per person using an equivalence scale or, more simply, income or consumption per capita. When family size falls and resources are held constant, income per person rises by construction, and demographic change may be recorded as income growth rather than as an independent driver of poverty decline. The resulting dividend echoes the macroeconomic demographic dividend, but the mechanism operates within families: fewer children share a given amount of household resources, raising resources relative to the poverty threshold. In the context of the global decline in fertility, accounts of antipoverty progress overstate the contribution of growth in household resources and understate the contribution of demographic change.

This resource dilution is the quantity–quality channel emphasized by Becker (1960) and Becker and Lewis (1973): an additional child mechanically lowers resources per person. Unless household resources rise enough to offset the larger denominator, the household moves closer to poverty. This is an accounting implication of per-capita measurement, not a claim that births or children reduce social welfare.

To quantify the family-size channel, I construct counterfactuals that replace the distribution of the number of co-resident children per mother in one survey with the distribution in another, and recalculate poverty rates holding all else fixed. Within specified cells, each mother in the base sample is reassigned the number of children at the same percentile of the target distribution. This rank-preserving transport (Novosad, Rafkin and Asher, 2022) changes family size and, with it, resources per person and population weights, while leaving household resources and other residents unchanged. I then recompute poverty under the counterfactual distribution.

The approach relies on a rank-preservation assumption: within cells, mothers retain their ordinal position in the co-resident child distribution as that distribution shifts across surveys. This assumption is weaker than assuming fertility is independent of resources. High-fertility mothers may be poorer, so long as their relative fertility ranks are stable as the distribution changes. Because the number of children is discrete, ranks within child counts are not observed. In the main specification, I form ranks within rural and urban cells and use random assignment to break ties. I also report results from a range of alternative cell groupings and tie-breaking rules that use the observed welfare–fertility copula, ascending and descending welfare ordering, and mother age. These alternative assignment rules produce tight ranges of counterfactual poverty rates, supporting the view that declining fertility contributed to the fall in global poverty.

The central estimate suggests that the fertility dividend accounts for 14.0% of the decline in all-person $3.00-per-day poverty, an even larger share (18.8%) of the decline in child poverty, and 16.7% of the decline among people in households with children. In countries where poverty rose, fertility decline often cushioned the increase. At higher poverty thresholds, the all-person share is 11.0% at $4.20 and 9.9% at $8.30. In richer Latin American countries, a larger share of the population lives near these higher thresholds, so a fixed denominator shift can have a larger effect there. In rapidly growing China, Indonesia, and Pakistan, much of the distribution has moved past the higher lines, so the same family-size shift crosses fewer households.

The results isolate the narrow, mechanical resource-dilution effect of fertility on poverty and thus likely represent a lower bound on the total causal effect of fertility change. In particular, this analysis deliberately excludes the effects of delayed childbearing and smaller families on parental labor supply, educational attainment, investments in children, or public budgets. To the extent those channels change observed household resources, this analysis attributes them to income or consumption growth.

The paper makes several contributions. First, it demonstrates that the decline in fertility and family size was a substantial contributor to the fall in global poverty. Because a country's fertility transition is a one-time event, this tailwind will fade. Second, it uses a rank-preserving transport method to decompose changes in poverty. The data cannot reveal which mothers would have had fewer children, so the paper reports ranges of estimates over a declared set of economically restricted rank assignments, in the spirit of partial identification (Manski, 2003; Tamer, 2010). Third, it applies these methods to a large array of countries and time periods. Finally, it extends the analysis to a longer horizon using IPUMS International censuses (Center, 2024). Because censuses lack comparable household welfare, these are one-directional implied effects at an observed survey's living standards, not observed poverty decompositions.

1.1 Relationship to the literature

The paper connects four literatures that usually proceed separately. First, the macroeconomic demographic-dividend literature studies how age structure affects aggregate saving, labor supply, and growth (Bloom and Williamson, 1998), and its simulation counterpart quantifies the growth effects of fertility reduction (Ashraf, Weil and Wilde, 2013). International poverty applications similarly relate fertility, dependency ratios, and poverty in macro or grouped data (Eastwood and Lipton, 1999; Cruz and Ahmed, 2018; Wietzke, 2020). Those estimates bundle many channels: the number of workers, age composition, household formation, human-capital investment, and general-equilibrium growth. I ask a narrower question, holding observed household resources fixed and measuring the direct change in poverty generated by the number of co-resident children sharing those resources. The calculation is an accounting statement about the poverty measure, not a model of development.

Second, poverty decompositions conventionally divide a change into growth and redistribution components (Datt and Ravallion, 1992; Ravallion, 2016). Because their welfare variable is already expressed per person or per equivalent adult, smaller families raise the welfare variable before the decomposition begins. The resulting change is therefore recorded as growth. World Bank applications add a demographic term by substituting the adult share or household-size ratio between surveys (Azevedo et al., 2013; Inchauste et al., 2014). Those substitutions operate on per-capita components country by country, bundle fertility with co-residence and aging, and do not preserve the joint distribution of family size and resources near the poverty line. The distinction in this paper is between growth in total household resources and growth in resources per person caused by a smaller denominator. Rather than re-decomposing the same scalar distribution, I use household rosters to open the welfare aggregate and transport one of its components.

Third, a large literature asks how household size and economies of scale should enter poverty measurement. Larger households may share housing, durable goods, fuel, and other public goods, while children may require fewer resources than adults (Lanjouw and Ravallion, 1995; Deaton and Paxson, 1998; Jolliffe and Tetteh-Baah, 2024). That literature mainly concerns poverty levels—who is poor under a per-capita, adult-equivalent, or scale-adjusted definition—while the question here concerns changes over time. Even if one accepts a particular scale, demographic change can move households across its threshold. The alternative-scale results therefore play two roles. They show that the official per-capita headline is implemented correctly, and they quantify how much of the historical result depends on the normative weight assigned to an additional child.

Fourth, recent work decomposes poverty changes using repeated cross-sections, cohorts, or panel transitions. Armentano, Niehaus and Vogl (2026), for example, asks how poverty fell by following changes within and across cohorts and reports little role for demographic covariates in its setting. That estimand differs from a distributional transport of children across the full resource distribution. A regression of within-household welfare changes on household-size changes conditions on the households observed in both periods and combines changes in resources, composition, and measurement. The transport here instead asks where the entire target fertility distribution would place households relative to a fixed poverty threshold. The two answers can coexist because they concern different populations and counterfactuals.

The analysis also relates to differential fertility. Fertility is not randomly distributed across the welfare distribution: poorer and less-educated women often have more children, and the gradient changes during development (Vogl, 2016). This selection is why the joint distribution matters: a shift of one child concentrated among households near the poverty line has a different headcount effect from the same marginal shift among the richest households. Rank-preserving transport retains the observed ordering while changing the marginal distribution, and the restricted-set analysis makes alternative economically motivated completions explicit.

Relative to these literatures, this paper measures the direct family-size component embedded in observed changes in per-capita poverty, rather than the effect of fertility on national income, the optimal equivalence scale, or the lifetime causal effect of an additional birth. The advantage of the approach is that the counterfactual stays close to the accounting identity, uses the full micro distribution, and applies consistently across countries; the cost is a transport assumption that repeated cross-sections cannot verify. The empirical design and robustness analysis are organized around that tradeoff.

2 Data and measurement

The analysis harmonizes household surveys with the World Bank Poverty and Inequality Platform (PIP; World Bank, 2026). The survey files supply household rosters, survey weights, urban or rural residence, and a household welfare aggregate. PIP supplies the official country-year headcount, reporting population, poverty line, and welfare concept. Each wave is anchored to its own country, survey year, and income or consumption concept. The analysis contains 79 survey waves and 57 adjacent pairs in 22 countries.

The study window is 1990–2019. It begins at the standard baseline for global poverty comparisons and ends before the COVID-19 disruption. When an adjacent survey pair straddles 1990 or 2019, both its observed poverty change and its transport effect are multiplied by the fraction of elapsed years inside the window. The proration is a linear interpolation within the boundary segment and admits no pre-1990 or post-2019 effects. The replication archive reports every pair's effect without boundary proration, so complete-pair sensitivity can be computed directly.

Household welfare follows PIP's concept for the corresponding survey. Anchoring rescales the welfare distribution to reproduce the official headcount while preserving within-wave ranks. The worst anchored-headcount discrepancy is 0.123 percentage points. Anchoring fixes the level of poverty, but it does not validate the density or ranking of households near the line. The paper therefore reports the fit of the transported child-count marginal, and the replication archive records the weighted mass near each line and every income-versus-consumption break. The worst target-distribution total-variation distance is 0.38%. These checks verify the implementation rather than the identifying assumption.

The main fertility measure is the number of co-resident children ages 0–17 linked to each co-resident mother. Exact mother pointers are used when both surveys in a pair contain reliable links. Otherwise, children are assigned to a plausible co-resident mother using a common probabilistic hierarchy. Children who cannot be linked are held fixed. Women with no linked co-resident children are not in the mother transport, so the estimand is the intensive distribution among observed co-resident mothers. This choice matches the resource-dilution question and avoids mistaking household recomposition for fertility, but it does not capture the extensive margin into motherhood.

The country sample is defined by data compatibility, not by the outcome. A survey wave enters when at least 80 percent of roster children can be linked to a co-resident mother, a rule fixed before any result is computed; it excludes a few early files whose rosters carry no usable mother pointers (Peru before 2004, Kenya 1992). All 22 countries with usable welfare, household, and mother-level roster information enter every headline result. The Philippines has a long survey series with under-15 counts but no mother link in the relevant files; its household-level result is reported in the appendix and is not mixed with the mother-level pool. Countries represented only in IPUMS enter the long-horizon demographic evidence, not the survey-welfare decomposition.

The 22 sample countries together held 81% of the world's $3.00-per-day poor in 1990, so the fixed sample covers most of the population whose poverty the global series tracks. The sample has two coverage limits, both reflecting data availability rather than design. First, the Chinese chain ends in 2002, the last CHIP survey with usable welfare and roster information; no comparable welfare microdata exist after that year, and IPUMS carries no post-2000 Chinese census sample. Most of China's poverty decline occurred after 2002, so the largest national episode in the global series is only partly covered. The truncation makes the cumulative count conservative, because post-2002 Chinese fertility decline receives no credit. The truncation's effect on the pooled share is less clear-cut: the cross-country pattern that family-size shares are smaller in fast-growth episodes suggests that extending the Chinese chain would, if anything, lower the pooled share somewhat rather than raise it. Second, India's NSS and HCES rosters record no mother line numbers, so every Indian pair relies on the probabilistic mother assignment; the rule can be validated against recorded pointers only in the surveys of other countries.

Figure 1 documents the economic selection underlying the design. In every sample country and wave, mothers in the poorest welfare quintile have more co-resident children than mothers in the richest quintile. The gradient generally compresses as fertility falls but never reverses. Rank preservation accommodates this selection; random assignment of fertility across welfare ranks would not.

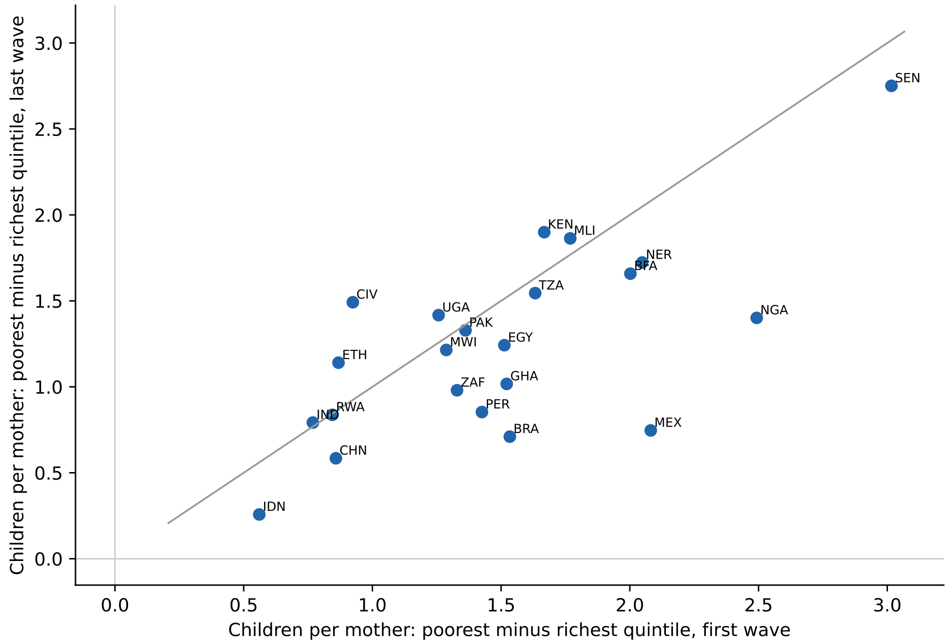

Figure 1: The welfare–fertility gradient persists as family size falls

Notes: Each point is a country. Axes show the survey-weighted difference in linked children per mother between the poorest and richest household-welfare quintiles in the first and last usable survey waves. The 45-degree line denotes an unchanged gradient.

3 Empirical method

Let household h have fixed resources R_h, fixed nonchildren A_h, and linked mothers m with child counts K_m, so that $K_h = \sum_{m \in h} K_m$. Under the official per-capita measure, welfare is

$$y_h = \frac{R_h}{A_h + K_h}.$$

Within urban/rural cell x, let $F_x^t(k)$ be the weighted distribution of children per linked mother in survey t. A mother at weighted rank p in the source distribution receives

$$K_m^{cf} = \min\{k : F_x^{t_1}(k) \geq p\}.$$

Counterfactual mother counts are summed within households, unlinked children and all nonchildren are held fixed, and poverty is recomputed from $R_h/(A_h + K_h^{cf})$. For child poverty, the household's poverty status is weighted by its counterfactual number of children; for the with-children outcome, the baseline set of households with children is retained.

The central counterfactual-assignment problem is how to attach the target family-size distribution to the source population of families and resources. The aggregate decline in family size is observed, but it does not by itself say whether the counterfactual change should be assigned to higher- or lower-resource families. I impose conditional rank preservation. Let u_m denote a mother's rank in her cell's resource distribution, v_m her latent fertility rank, and C_x^t their joint distribution within urban/rural cell x in survey t. The identifying condition is stability of that dependence structure across adjacent surveys:

$$C_x^{t_0}(u, v) = C_x^{t_1}(u, v).$$

In plain terms, mothers who tend to have more children than their peers in one era would also tend to have more children in another, and they occupy similar positions in the resource distribution. The condition allows fertility to be correlated with resources and observed characteristics. It rules out reranking of latent fertility propensities against resources between adjacent surveys. Because child counts are discrete, an observed count reveals the latent fertility rank only up to a mass point, so the within-tie assignment is genuinely unidentified; Section 5 reports the declared set of admissible completions.

For each adjacent pair and each population $j \in \{all, child, with\text{-}child\}$, let D_F^j be the source-wave poverty reduction from imposing the later child distribution, and let D_R^j be the signed later-wave effect of imposing the earlier distribution. The pair effect is

$$D^j = \tfrac{1}{2}(D_F^j + D_R^j).$$

The forward calculation prices demographic change at the earlier welfare distribution; the reverse calculation prices it at the later distribution. Averaging is a symmetric index-number solution, and I apply it to all three outcomes, so differences across the all-person, child, and with-children estimates reflect their population weights rather than different directions or survey samples. Consecutive pair effects are then summed into country chains. One qualification applies to the with-children outcome: each leg conditions on its own baseline set of households with children, and fertility decline itself moves households out of that set, so this estimand averages over the early and late with-children populations rather than a single fixed set.

The pooled share is constructed from counts. For country-pair q, population N_q^j, exposure fraction s_q, transport effect D_q^j, and observed change ΔH_q^j, the pooled statistic is

$$S^j = \frac{\sum_q s_q N_q^j D_q^j}{\sum_q s_q N_q^j \Delta H_q^j}.$$

This formula uses all compatible countries, including negative ΔH_q^j. The numerator is a cumulative sum of period-specific headcount contributions, not a count of the unique people who ever crossed the line.

Alternative equivalence scales replace $A + K$ consistently in both the actual and counterfactual poverty measures and each wave is reanchored to its PIP headcount. An earlier implementation reanchored the actual distribution under an alternative scale but evaluated the counterfactual with the per-capita denominator, mechanically understating differences across scales. The corrected specifications use the same scale on both sides. I report the square-root, Betson (Betson, 1996), OECD-modified, and adults-plus-half-a-child scales. Because these scales give an additional child less than unit need, they should produce smaller effects than the official per-capita measure.

4 Results

Table 1 reports the headline results. At \$3.00 per day, smaller families account for 14.0% of the pooled all-person decline, 18.8% of the child-poverty decline, and 16.7% of the decline among people in households with children. The child and with-children estimates are larger because the affected population receives greater weight, not because the transport differs. The cumulative all-person numerator is 160 million, best interpreted as the sum of poor-count contributions over country-periods. The pooled denominator is the net change across all countries, including the 6 countries where poverty rose; restricting the pool to countries with observed declines gives 13.5%, and to countries with declines above 10 percentage points gives 13.3%.

Table 1: Family-size contribution on a fixed all-country sample

Poverty line	All persons	Children	With-child HH	Countries
\$3.00	14.0%	18.8%	16.7%	22
\$4.20	11.0%	16.3%	13.9%	22
\$8.30	9.9%	18.2%	14.7%	22

Ratios are summed population-count transport effects divided by summed observed population-count declines. All outcomes average forward and reverse transports for every adjacent pair and sum the same pair chain. No country is selected on the sign or size of its poverty change.

Figure 2 shows every country in percentage points. Gray bars are the observed all-person poverty decline, so bars to the left of zero identify countries where poverty rose. Colored symbols show the family-size effects. Fertility decline reduced poverty in most countries even when aggregate poverty rose, acting as partial insurance against crises or stagnant household resources. Including these countries prevents selection on the denominator of the pooled share.

The cross-country pattern separates the mechanism from the macro environment. In fast-growth episodes, the family-size effect can be sizable in percentage points but a modest fraction of a very large decline. In crisis or low-growth episodes, the same demographic movement can be the principal poverty-reducing force. A single worldwide ratio therefore conveys aggregate scale but obscures this variation.

Figure 3 addresses poverty-line heterogeneity. Effects need not be monotone in the line: they depend on the density of households close enough to the threshold for a denominator change to alter poverty status. In Mexico, Brazil, and Peru, family-size effects are at least as large at \$4.20 and \$8.30 as at \$3.00. That pattern is consistent with the intuition from richer

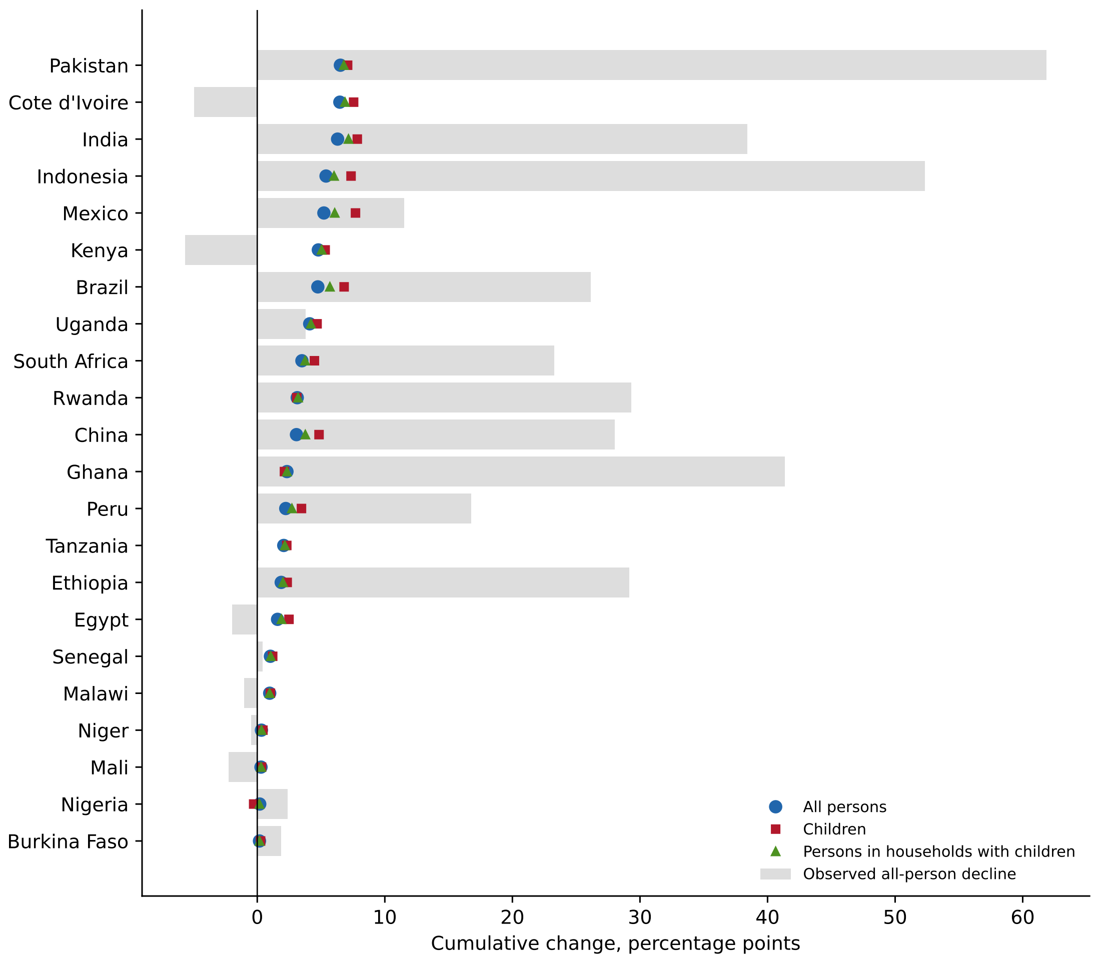

Figure 2: Family-size effects in all compatible countries

Notes: Gray bars show cumulative observed all-person changes; positive values are poverty declines. Symbols show symmetric family-size effects for the same adjacent pairs. Countries are not selected on the sign or magnitude of the observed change.

economies: extreme poverty can already be rare while many households remain near a higher, nationally salient threshold. In China and Indonesia, the higher-line effect is smaller over the observed window because rapid growth moved much of the welfare distribution through those thresholds. The pooled fixed-sample shares are 14.0%, 11.0%, and 9.9%, respectively.

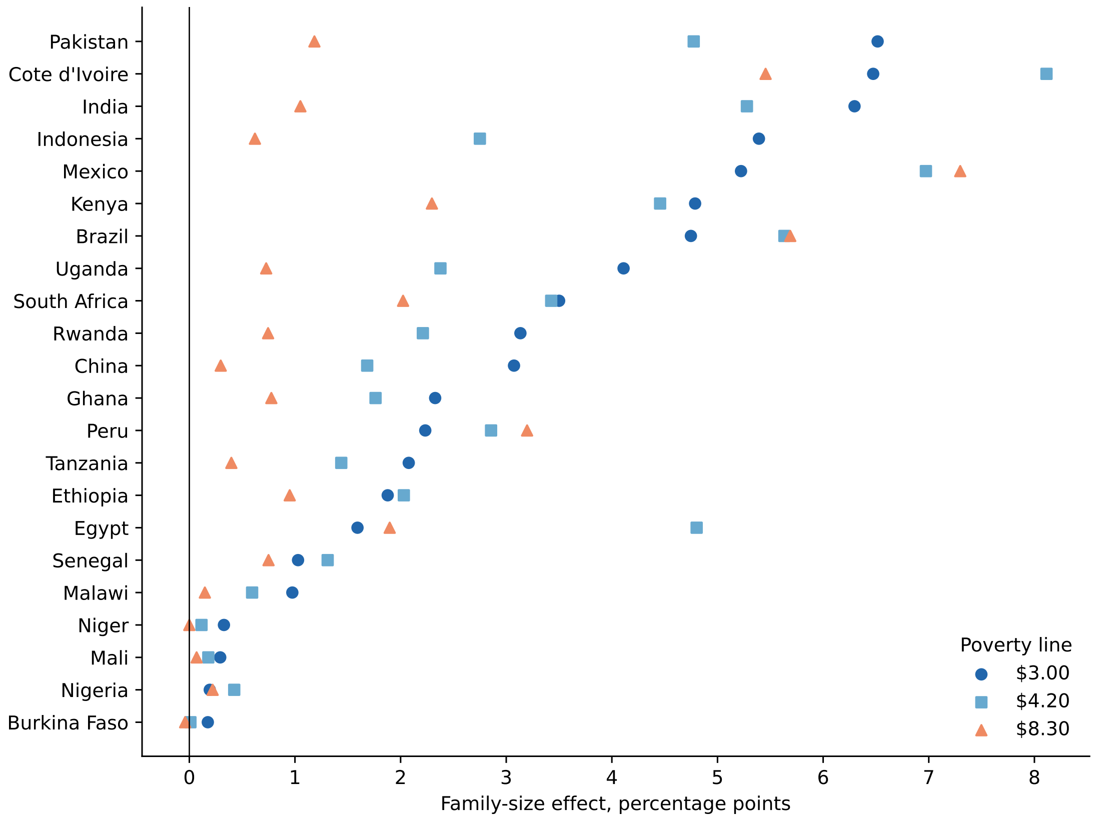

Figure 3: The poverty-line response differs across development paths

Notes: All-person percentage-point effects use the same countries, survey pairs, symmetric estimator, and 1990–2019 exposure at each line. Each wave is reanchored to PIP at the indicated line.

Standard growth–redistribution decompositions place this channel in the growth component. They compare welfare per person across surveys after household size has already entered the welfare aggregate. The transport instead holds total household resources fixed and changes only the number of claims on those resources. The comparison shows that the "growth" term in conventional decompositions combines growth in household resources with a demographic denominator.

4.1 Why the magnitude varies across countries

Three features govern the country effect. The first is the size and timing of the fertility shift. A long survey chain spanning a large fall in children per mother offers more scope for a denominator contribution than a short chain late in a stalled transition. Coverage clocks and survey windows therefore matter: a country with a substantial national fertility transition can have a modest survey estimate if its welfare microdata begin after most of the decline.

The second feature is the welfare–fertility gradient. If large families are concentrated among poorer households, transporting the later, smaller-family distribution can move more households across a low poverty line. The relevant quantity, however, is not the unconditional correlation alone but the assignment of changes within child-count ranks and urban/rural cells, particularly among households near the threshold. The gradient plots and the rank-completion set carry more of this information than a single regression coefficient.

The third feature is the density of welfare near the poverty line. A mechanical change in resources per person affects the headcount only when it changes an indicator. If few households lie near a line, even a large average denominator change has little headcount effect. Conversely, a smaller demographic change can produce a large poverty effect when the welfare distribution is compressed around the line. Baseline poverty rates alone therefore do not order the estimates, and the response can rise with the threshold in some middle-income countries.

These forces also clarify why effects and shares differ. An effect in percentage points describes the counterfactual change generated by fertility at the observed welfare distributions. A share divides that effect by the observed poverty decline, which may be large, small, zero, or negative for reasons unrelated to fertility. Shares are useful for aggregate historical attribution, but they are unstable in crisis episodes. The main country figure therefore leads with percentage points and the pooled share is constructed only after summing population counts.

5 Identification, robustness, and long-horizon evidence

Figure 4, panel A, reports economically restricted sets rather than unrestricted Fréchet–Manski bounds (Manski, 2003). The latter permit any coupling of target fertility with source resources and therefore span assignments that reverse the stable, negative welfare–fertility gradient documented in Figure 1. They are useful for showing that rank assumptions carry identifying content, but not for measuring a plausible fertility dividend. The reported set admits four completions of discrete ranks: random completion; a Gaussian copula calibrated to the observed welfare–fertility association; mother-age ordering, which allows fertility timing to structure ranks; and monotone selection in which poorer mothers have weakly higher latent fertility propensity within a child-count mass point. Across these restrictions, the all-person share is 13.6%–14.7%, the child share 18.8%–20.4%, and the with-children share 16.4%–17.6%.

The sets are assumption-indexed. They vary only the assignment within observed child-count ties; rank preservation itself is held fixed, so the width of the range measures sensitivity to the tie completion rather than to the identifying assumption, which repeated cross-sections cannot verify directly. The ranges are not sampling confidence intervals, and I do not claim sharpness under every possible economic model. They replace arbitrary worst-case couplings with restrictions motivated by persistent differential fertility, life-cycle timing, and observed rank dependence. Cells that combine urban status with mother age are not used: the transport is defined within urban/rural cells, and mother age enters instead through the age-ordered tie completion.

A direct stress test perturbs rank preservation itself rather than the tie completion. Each mother's latent transport position is a mixture of her observed fertility rank and an independent random draw, with mixing weight λ; $\lambda = 1$ assigns the target family-size marginal at random within urban/rural cells and so eliminates rank preservation entirely. The pooled all-person share is 14.1% at $\lambda = 0$ (this reproduces the headline ordering up to an independent set of random within-tie completion draws, which accounts for the small gap from

14.0%), 14.2% at $\lambda = 0.25$, 14.4% at $\lambda = 0.5$, and 14.9% under fully random assignment. The estimate is stable under reranking. If anything, rank preservation is conservative: keeping poorer mothers at high fertility ranks concentrates the remaining children among poor households, so random assignment yields a slightly larger family-size contribution.

Panel B reports the IPUMS extension. For a census year without comparable welfare, I borrow an observed survey's welfare distribution and welfare–fertility rank mapping, impose the census children-per-mother marginal, and recompute poverty. The result measures how much higher or lower poverty would have been at the anchor survey's living standards under the census year's family-size distribution. Because the census observes no comparable welfare, there is no reverse transport to average, so these one-directional estimates are neither inserted into the symmetric survey chain nor pooled into the 1990–2019 headline. A negative implied effect means the census children-per-mother distribution lies below the anchor survey's, so poverty at the anchor's living standards would have been lower under census family sizes. For the pre-survey censuses in the Sahel, this signals co-resident family sizes that rose toward the anchor. That pattern foreshadows the survey-window finding that those countries' recent fertility declines mainly cushioned rising poverty.

The IPUMS exercise uses exact census mother pointers and covers pre-survey, interior, and post-survey census years where available. Reliability is determined from census link quality and, where a census falls within three years of a survey wave, from fertility agreement at the overlap; countries with no such near-overlap pass on link quality alone. The gate never uses the magnitude or sign of the projected effect. 14 countries pass the prespecified gate. Across 20 census–survey overlaps within two years, the median absolute children-per-mother difference is 0.10 and the median absolute implied poverty effect is 0.59 percentage points. The appendix compares nearest and distant survey anchors. Divergence between them measures dependence on the borrowed welfare distribution and is reported rather than averaged away.

Conditional household bootstrap intervals are 13.7%–14.3% for all persons, 18.4%–19.3%

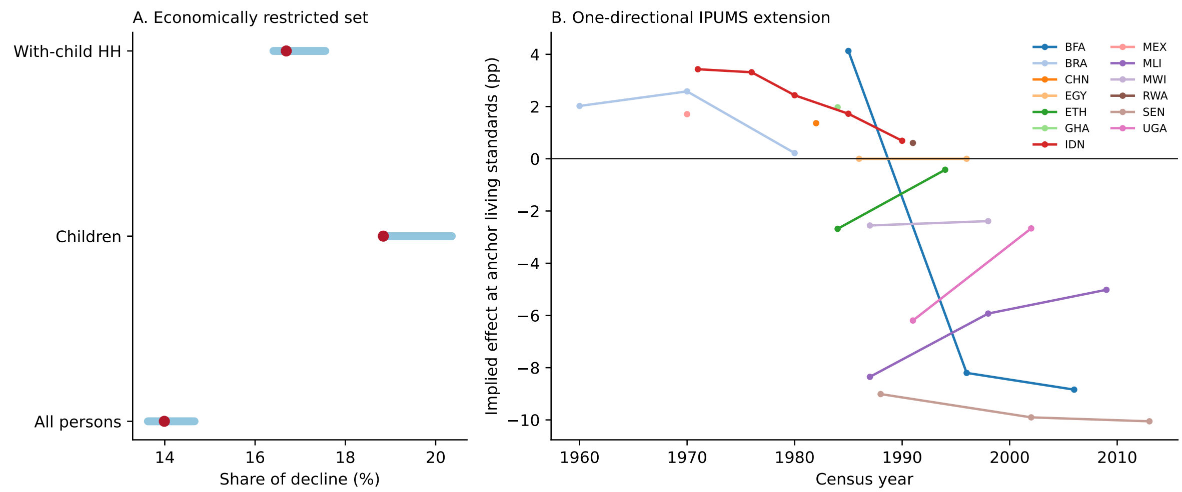

Figure 4: Economically restricted assignments and one-directional census extensions

Notes: Panel A shows the range over the four declared economically admissible within-tie completions; red dots are random-completion estimates. Panel B imposes IPUMS census fertility marginals on the nearest observed survey welfare distribution for every reliable country in every non-overlap census year. These implied effects are not observed poverty changes.

for children, and 16.3%–17.1% for people in households with children. The resampling uses common Poisson household weights for repeated appearances of a survey wave in adjacent pairs, while holding PIP anchors and the random transport completion fixed. Because several harmonized files do not retain strata and PSU identifiers, these intervals are not design-aware, and holding the anchors and assignments fixed conditions them on the estimated transport map as well, so they understate total sampling uncertainty. They quantify conditional sampling variation; identification sensitivity remains the more important source of uncertainty.

Equivalence scales change the answer for a clear economic reason. Relative to the official per-capita estimate of 14.0%, the square-root scale yields 7.1%, Betson 6.5%, OECD-modified 8.6%, and adults-plus-half-a-child 9.2%. The smaller estimates reflect the scales' assignment of less additional need to a child, not a different mechanism. Allowing household resources to fall by 5 or 10 percent for each removed child lowers the estimate to 10.0% and 5.8%. Those offsets are behavioral scenarios, not accounting identities; they bracket the possibility that larger families produce additional income or scale economies. Measured child income shows the scenarios are generous: in two sample surveys that record person-level income by source, all members under 18 combined receive 3.5% of household income among Mexican households with children and 1.6% in Brazil—less than the 5 percent the milder scenario assumes for each removed child (appendix).

5.1 Threats to interpretation

Five limitations qualify the claim. First, repeated surveys do not observe the same mothers. Rank preservation is a distributional invariance assumption, not an individual fixed effect. Migration, mortality, fostering, and changing definitions of the household can violate it. Adjacent-pair chaining reduces the distance over which invariance is imposed, and forward/reverse averaging reduces index-number dependence, but neither solves unobserved reranking.

Second, co-resident children per mother is not completed fertility. It combines births with child survival and residence. This is the correct roster measure for the resources actually shared within a surveyed household, but it should not be labeled a pure biological-fertility effect. Census children-ever-born measures provide a useful demographic check while answering a different question.

Third, holding resources fixed is deliberately partial equilibrium. If an additional child changes parental labor supply, transfers, remittances, or production, the total causal effect can differ. The resource-offset scenarios show how large a proportional response would need to be to attenuate the accounting result; they do not estimate that response. Fourth, PIP anchoring cannot repair errors in welfare ranks or survey comparability. Concept breaks and near-line densities are recorded in the replication archive. Fifth, conditional household bootstraps understate total uncertainty when survey-design variables are unavailable. Publication of the final archive should preserve strata and clusters wherever licenses allow.

The estimates are therefore best read as direct effects of observed family-size change on measured poverty under economically restricted distributional assignments, not as model-free bounds or estimates of the welfare value of population change.

6 Conclusion

Declining family size was a substantial and overlooked contributor to the fall in global absolute poverty. On a fixed sample of all compatible countries, smaller families account for 14.0% of the all-person decline at $3.00 per day, with larger contributions for children and people in households with children. The mechanism is direct resource dilution: fewer children divide a given amount of household resources, raising resources per person. Because conventional poverty accounting conditions on family size, it records this demographic improvement as income or consumption growth.

The analysis is deliberately narrower than a total causal effect of fertility. Holding household resources fixed excludes parental labor supply, human capital, child quality, timing, and public finance. It also does not value births or the welfare of people who do not appear in a counterfactual population. The result is a decomposition of measured poverty under explicit transport assumptions.

Three features make the international evidence especially informative. The effect appears in fast-growth, crisis, and stalled-transition settings; it varies across poverty lines with the density of households near each threshold; and it extends backward and forward in IPUMS censuses when interpreted at observed anchor living standards. No single country, poverty line, or survey series drives the result.

One practical implication follows. A demographically standardized poverty series, holding a base children-per-mother distribution fixed, could accompany the conventional headcount. Such a series would separate growth in household resources from changes in the number of claims on those resources and make visible a channel that official per-capita statistics combine with growth.

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Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

Code, generated results, tests, and provenance records are included in the replication archive. Licensed household microdata cannot be redistributed. Public PIP and IPUMS inputs are documented by source and vintage, and every numerical statement and exhibit is generated from the analysis outputs.

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A Country coverage and survey chains

Table 2: All compatible country chains at $3.00 per day

Country	Window	Observed (pp)	Effect (pp)	Share*	Exact/pairs
Burkina Faso	2018-2019	1.9	0.2	9.3%	0/1
Brazil	1990-2019	26.1	4.7	18.2%	1/3
China	1990-2002	28.0	3.1	11.0%	0/2
Cote d'Ivoire	1990-2019	-5.0	6.5	–	0/2
Egypt	1999-2019	-2.0	1.6	–	7/7
Ethiopia	1999-2019	29.2	1.9	6.4%	1/3
Ghana	1991-2016	41.3	2.3	5.6%	1/2
Indonesia	1993-2014	52.3	5.4	10.3%	4/4
India	1990-2019	38.4	6.3	16.4%	0/4
Kenya	1994-2015	-5.7	4.8	–	0/3
Mexico	1990-2019	11.5	5.2	45.4%	1/4
Mali	2018-2019	-2.3	0.3	–	0/1
Malawi	2004-2019	-1.0	1.0	–	0/1
Niger	2018-2019	-0.5	0.3	–	0/1
Nigeria	2010-2019	2.4	0.2	8.1%	1/2
Pakistan	1991-2019	61.9	6.5	10.5%	3/6
Peru	2004-2019	16.8	2.2	13.3%	0/3
Rwanda	2000-2019	29.3	3.1	10.7%	0/2
Senegal	2018-2019	0.4	1.0	245.1%	0/1
Tanzania	2011-2017	0.1	2.1	2115.3%	0/1
Uganda	2009-2019	3.8	4.1	108.6%	0/1
South Africa	1993-2019	23.3	3.5	15.1%	0/3

*A share is suppressed when poverty rose or was unchanged because the ratio is not interpretable; every row remains in the pooled count calculation.

All compatible mother-level chains remain in the fixed sample. Shares are omitted for country rows in which poverty rose because a positive poverty-reducing effect divided by an observed increase in poverty is not informative. Those observations remain in every population-count pool.

The 1990–2019 boundary rule prorates a pair only when it crosses an endpoint. Boundary-prorated pairs contribute 34.6% of the cumulative all-person count numerator, so the linear-accrual assumption matters for the aggregate magnitude. Complete-pair results and alternative windows can be computed directly from the unprorated per-pair effects in the replication archive. Concept breaks are retained in the headline only when they are part of PIP's own country series; within-concept chains and Datt–Ravallion comparators are reported as diagnostics and are never selectively used to replace the primary survey pairs.

Table 3: Equivalence-scale and resource-offset specifications

Specification	All	Child	With-child HH
Per capita (official)	14.0%	18.8%	16.7%
Square root	7.1%	9.4%	8.5%
Betson	6.5%	9.1%	7.9%
OECD modified	8.6%	12.1%	10.4%
Adults + 0.5 child	9.2%	13.3%	11.1%
5 percent resource offset	10.0%	13.0%	11.7%
10 percent resource offset	5.8%	6.7%	6.3%

B Equivalence scales and resource responses

For every alternative scale, the actual and transported household denominators use the same formula, and every wave is reanchored to its PIP headcount. The Betson scale is $(A + .5K)^{.7}$ for two or more adults and $(1.3 + .5K)^{.7}$ for one adult. Resource offset scenarios multiply resources by $(1 - o)^{K - K^{cf}}$ and are symmetric in forward and reverse calculations. Counting dependent children as ages 0–14 rather than 0–17, with the same pairs and count-weighted pooling, yields 11.7% for all persons, 17.7% for children, and 15.2% for people in households with children.

Measured child income is far smaller than either offset scenario. In the ENIGH and PNADC files, which record income by source for each household member, all members under 18 combined receive 3.5% of household income among Mexican households with children in 2016 (3.1% in 2024) and 1.6% among Brazilian households with children in 2023; including childless households lowers the shares to 2.0% and 0.7%. A 5 percent offset for each removed child therefore assumes a larger income response than the combined measured contribution of all resident children. Children in these settings consume far more than they earn, and the fixed-resource baseline sits closer to the measured data than either scenario.

C Restricted identification set

The main paper reports the finite union of four economically motivated completions of latent ranks within observed child-count mass points. Random completion imposes no additional within-tie ordering. The copula completion calibrates dependence to the observed normal-score correlation of welfare and fertility. Age completion orders latent ranks by mother age. The monotone-selection completion assigns poorer mothers weakly higher latent fertility propensity, consistent with the sign of the observed cross-sectional gradient. The minimum and maximum over these declared completions form the reported restricted set. It is not the unrestricted Fréchet–Manski set, and I do not claim sharpness under models outside these restrictions. The completions vary only the within-tie assignment; the rank-preservation condition itself is not varied. The reverse ordering, in which richer mothers have higher latent fertility within a tie, is computed as a diagnostic but excluded from the set because the observed welfare–fertility gradient is negative in every sample country and wave (Figure 1).

D IPUMS International extension

Table 4: Earliest reliable one-directional IPUMS extension by country

Country	Census	Anchor	Relation	Nearest (pp)	Distant (pp)
Burkina Faso	1985	2018	pre	4.13	5.33
Brazil	1960	1981	pre	2.02	4.33
China	1982	1988	pre	1.36	5.43
Egypt	1986	1999	pre	0.00	1.84
Ethiopia	1984	1999	pre	-2.68	2.47
Ghana	1984	1991	pre	1.97	5.33
Indonesia	1971	1993	pre	3.43	12.80
Mexico	1970	1984	pre	1.71	3.55
Mali	1987	2018	pre	-8.35	-9.15
Malawi	1987	2004	pre	-2.55	0.45
Peru	2007	2004	interior	-1.99	–
Rwanda	1991	2000	pre	0.61	7.83
Senegal	1988	2018	pre	-9.01	-6.64
Uganda	1991	2009	pre	-6.19	-1.30

The table displays the earliest reliable census for each country. The replication CSV reports every reliable census and all three poverty lines. Distant-anchor estimates use the survey wave farthest in time from the census and are sensitivity checks, not estimates to be averaged; a dash marks censuses whose farthest wave lacks usable mother links.

The nearest-anchor estimate holds fixed the closest survey welfare distribution. The distant-anchor column repeats the exercise using the survey wave farthest in time from the census; anchor waves may include surveys outside the 1990–2019 estimation window. Large differences are evidence that the implied historical poverty effect is sensitive to the borrowed welfare distribution. No forward/reverse average is formed because the census does not observe comparable welfare. Mother age was not retained in the census marginal files, so the census transports are not age-standardized.

E Additional data that are not pooled

The Philippine FIES series observes under-15 child bands but not mother pointers in the historical welfare files. Household-level under-15 transports are reported as a separate sensitivity series because pooling them with children-per-mother would change the estimand. Demographic-only IPUMS countries likewise enter the fertility panel but not the welfare decomposition. This separation uses all of the data while keeping incompatible estimands apart.

Table 5: Household-level under-15 fallback not included in the mother-level pool

Country	Window	Observed decline (pp)	Effect (pp)	Share
Philippines	1990–2009	17.3	6.0	34.9%

This transport uses under-15 children per household because the historical FIES files do not contain mother pointers. It is reported to use the available welfare series but is not pooled with children-per-mother estimates.